

THEORY OF CHANGE: REDUCING AIR POLLUTION IN METRO CITIES





Theory of Change: Reducing Air Pollution in Metro Cities

Building Cleaner, Healthier and Sustainable Urban Environments through Integrated Air Quality Management

Prepared by: Vishwajeet Kumar



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Context & Problem Statement

Air Quality Crisis

PM2.5 levels 2–3× WHO guidelines in major metro cities – a systemic public health emergency.

Primary Emission Sources

Vehicular emissions 40% · Industrial 30% · Construction dust 20% · Open burning 10%

Health Impact

1.2 million premature deaths annually in India attributable to air pollution exposure.

Enforcement Gap

Weak compliance mechanisms and underdeveloped monitoring infrastructure compound the crisis.



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Stakeholder Mapping



Primary Beneficiaries

Urban residents, vulnerable groups including children and elderly, and low-income communities bearing disproportionate health burdens.

Government Actors

MoEFCC, CPCB, SPCBs, Municipal Corporations, and Transport Departments provide regulatory authority and oversight.

Implementing Agencies

Pollution Control Boards, Municipal Authorities, and Public Transport Agencies drive on-ground execution.

Partners

Industries, EV manufacturers, renewable energy providers, NGOs, and academia enable innovation and community engagement.



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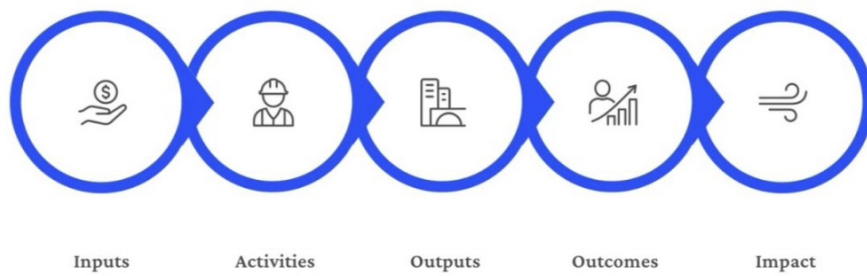
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Theory of Change Framework

A backward-mapped causal chain linking strategic inputs to transformational impact — from resources mobilised to communities protected.



Each causal link is underpinned by explicit assumptions, evidence-based interventions, and measurable indicators to ensure accountability at every stage.



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Inputs: Resources & Enablers



Financial

Government funding, green bonds, climate finance, and private sector investment mobilised at scale.



Human Capital

Skilled workforce, technical experts, M&E teams, and research institutions providing evidence-based guidance.



Infrastructure

Air quality monitoring stations, digital management systems, and expanded transport networks.



Policy Levers

NCAP guidelines, BS-VI standards, regulatory frameworks, and robust enforcement mechanisms.



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Activities: Programme Interventions



Transport

Expand public transit networks, develop EV charging infrastructure, and promote clean fuel adoption.



Industrial

Strengthen emission monitoring, enforce compliance, and support clean technology adoption by industry.



Urban Management

Implement construction dust control, prevent open waste burning, and drive urban afforestation.



Engagement

Public awareness campaigns, inter-agency coordination mechanisms, and community participation platforms.



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Outputs: Immediate Deliverables



Infrastructure

EV charging stations established across metro corridors; public transport routes and frequency expanded.



Compliance

Industrial emission standards implemented and enforced; targeted enforcement drives conducted city-wide.



Monitoring

Air quality stations strengthened across cities; real-time AQI data systems made publicly operational.



Awareness & Green Spaces

Public campaigns completed; green cover expanded; dust control measures activated at construction sites.



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Outcomes: Short, Medium & Long-term



Short-term

Increased public awareness, improved regulatory compliance, and accelerated adoption of cleaner transport options.



Medium-term

Measurable reduction in industrial and vehicular emissions; strengthened air quality management systems across agencies.



Long-term

Sustained PM2.5/PM10 reduction to safe levels; improved respiratory health outcomes; enhanced urban sustainability.



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Impact: Vision & SDG Alignment

Primary Impact

Cleaner air, healthier communities, and climate-resilient metro cities where all residents breathe safely.

Health Outcomes

Reduced pollution-related mortality, improved life expectancy, and measurably enhanced quality of life across all income groups.

SDG Alignment

SDG 3 Good Health · **SDG 11** Sustainable Cities · **SDG 13** Climate Action



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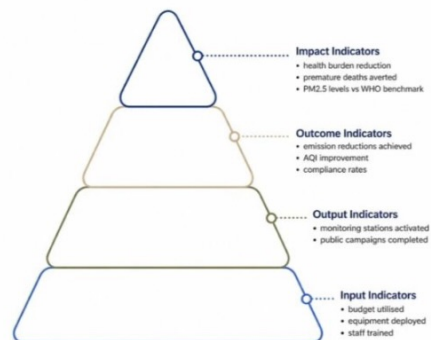
Assumptions, Risks & Monitoring Framework

Key Assumptions

- Strong and sustained political commitment at national and city levels
- Adequate funding flows maintained across programme lifecycle
- Active public participation and inter-agency coordination

Critical Risks

- Weak enforcement capacity and regulatory capture
- Funding delays and fiduciary bottlenecks
- Low behavioural change adoption and industrial resistance



Programme success hinges on adaptive management – regularly reviewing assumptions, tracking indicators, and course-correcting based on real-time AQI and health data.



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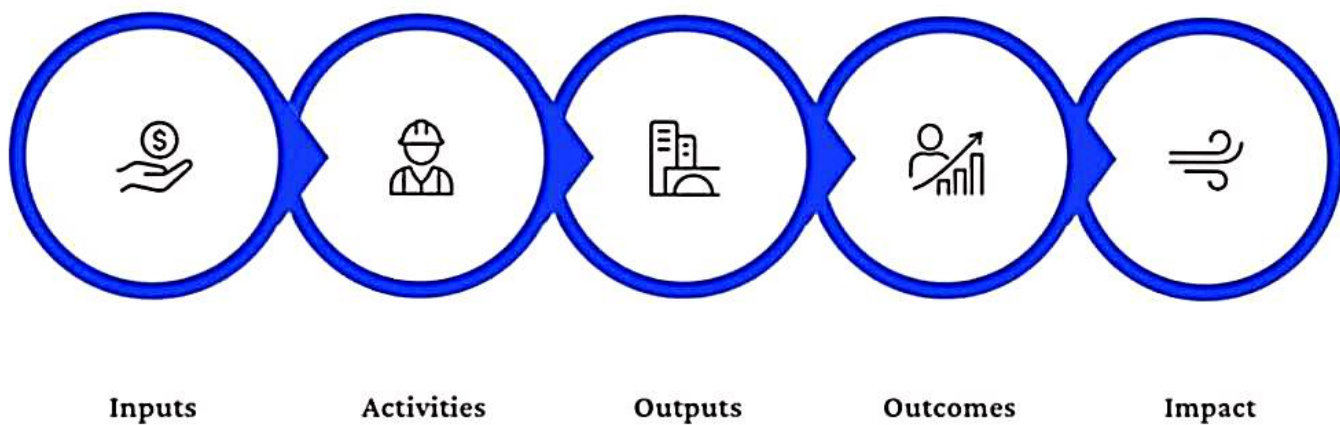
Stakeholder Mapping



- Primary Beneficiaries**
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MoEFCC, CPCB, SPCBs, Municipal Corporations, and Transport Departments provide regulatory authority and oversight.
- Implementing Agencies**
Pollution Control Boards, Municipal Authorities, and Public Transport Agencies drive on-ground execution.
- Partners**
Industries, EV manufacturers, renewable energy providers, NGOs, and academia enable innovation and community engagement.

Theory of Change Framework

A backward-mapped causal chain linking strategic inputs to transformational impact – from resources mobilised to communities protected.



Each causal link is underpinned by explicit assumptions, evidence-based interventions, and measurable indicators to ensure accountability at every stage.

Inputs: Resources & Enablers



Financial

Government funding, green bonds, climate finance, and private sector investment mobilised at scale.



Human Capital

Skilled workforce, technical experts, M&E teams, and research institutions providing evidence-based guidance.



Infrastructure

Air quality monitoring stations, digital management systems, and expanded transport networks.



Policy Levers

NCAP guidelines, BS-VI standards, regulatory frameworks, and robust enforcement mechanisms.

Activities: Programme Interventions



Transport

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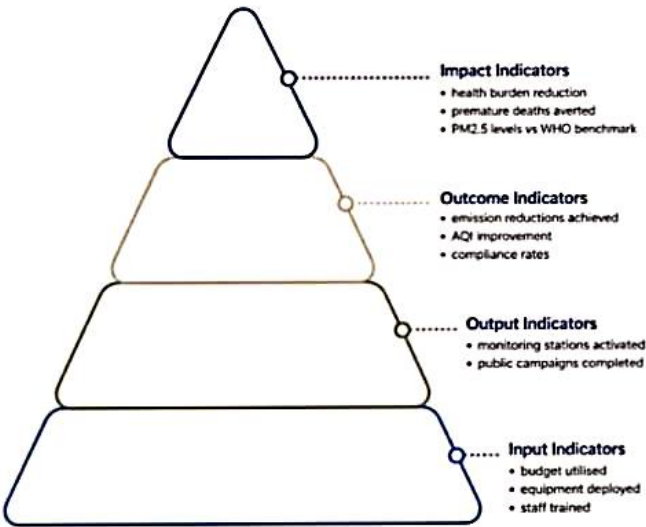
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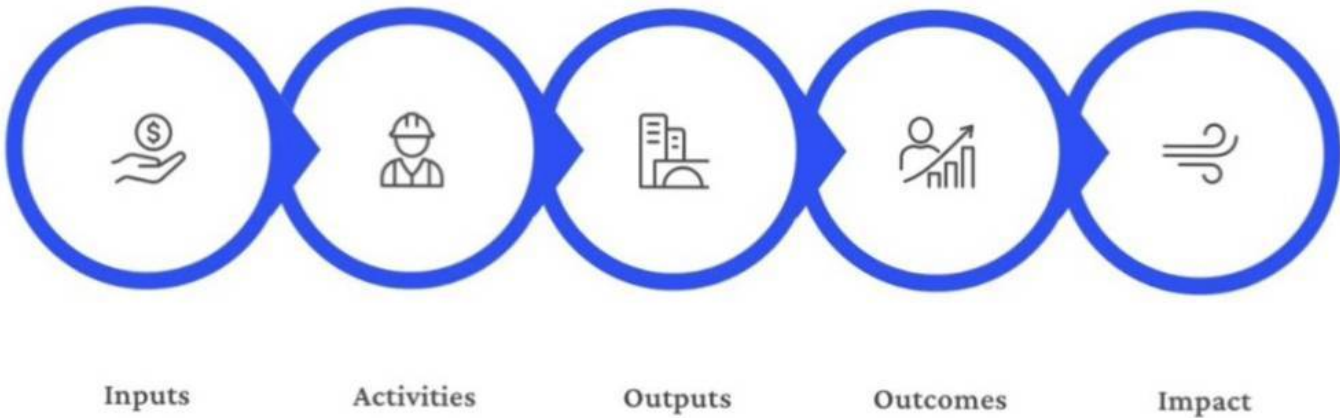
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